SANOFI

COMPREHENSIVE ORGANIZATIONAL OVERVIEW

Data Functions, Business Operations & Analytics

Chasing the Miracles of Science

Date: 11 February 2026

Prepared by: Atukuri Maithili

Department: Go-To-Market Capabilities (GTMC) - Insights & Analytics (I&A) - Global Data Management

Role : Intern – Data and process management

INTRODUCTION

Document Purpose

This comprehensive organizational overview provides a detailed examination of Sanofi's structure, operations, and data ecosystem. The document serves as a strategic reference guide for understanding how Sanofi leverages data functions, business operations, and analytics to support its mission of "Chasing the Miracles of Science."

Scope and Coverage

This document encompasses:

- Organizational Structure: Complete overview of Sanofi's Global Business Units (GBUs) and Global Functions
- Business Operations: Detailed analysis of Go-To-Market Capabilities (GTMC) and operational frameworks
- Data Functions: Comprehensive examination of data services, governance, and analytics capabilities
- Systems and Technology: Overview of enterprise systems, databases, and platforms supporting operations
- Data Quality Management: Detailed framework for ensuring data accuracy, completeness, and reliability
- Stakeholder Ecosystem: Mapping of internal and external data providers and consumers
- Performance Management: Key Performance Indicators (KPIs) and metrics for operational excellence

TABLE OF CONTENTS

1. SANOFI OVERVIEW 7
    - 1.1 Mission 7
    - 1.2 Organizational Structure at Sanofi 7
        - 1.2.1 Global Business Units (GBUs) 7
        - 1.2.2 Sanofi Global Functions 8
    - 1.3 Core Therapeutic Areas 7
2. CORPORATE STRUCTURE 9
    - 2.1 Executive Committee 9
    - 2.2 Major Divisions 9
3. GO-TO-MARKET CAPABILITIES (GTMC) UNDER BUSINESS OPERATIONS 9
    - 3.1 GTMC Executive Leadership 9
    - 3.2 GTMC Hyderabad Hub Structure 9
        - 3.2.1 Insights & Analytics (I&A) - Hub 9
        - 3.2.2 Customer Facing Operations (CFC) - Hub 10
        - 3.2.3 Omnichannel & LEX/MEX - Hub 10
4. INSIGHTS AND ANALYTICS (I&A) 10
    - 4.1 Overview 10
    - 4.2 I&A Core Teams 10
        - 4.2.1 Business Insights and Analytics (BI&A) 10
        - 4.2.2 Commercial Data Management 11
        - 4.2.3 Visual Analytics and Data Visualization 11
        - 4.2.4 Advanced Data Analytics and Processing Technology (ADAPT) ... 11
        - 4.2.5 Sales Force Effectiveness (SFE) Analytics 11
        - 4.2.6 Market Intelligence and Competitive Analysis 11
        - 4.2.7 Customer Analytics 11

- 4.2.8 Forecasting and Predictive Analytics 12
- 4.3 I&A Value Proposition 12

5. SYSTEMS, DATABASES, AND PLATFORMS 12
    - 5.1 Enterprise Databases 12
        - 5.1.1 On-premise 12
        - 5.1.2 Cloud databases 12
        - 5.1.3 Specialized systems 12
    - 5.2 Operational and Commercial Systems 12
    - 5.3 System Data Sources 13
6. DATA FOUNDATIONS 13
    - 6.1 ADAPT (Regional Commercial Data and Analytics Platform) 13
        - 6.1.1 Definition 13
        - 6.1.2 Architecture 14
        - 6.1.3 Regional Variants 14
        - 6.1.4 Data Domains 14
        - 6.1.5 Benefits 14
    - 6.2 MSDF (Market and Sales Data Foundation) 15
    - 6.3 Advanced Analytics Platforms 15
7. TYPES OF DATA MANAGED BY GTMC AND I&A 15
    - 7.1 Commercial Data 15
    - 7.2 Market Intelligence Data 15
    - 7.3 Customer Insights Data 15
    - 7.4 Operational Data 16
    - 7.5 Financial Data 16
8. END-TO-END DATA FLOW AND OPERATING MODEL 16
    - 8.1 Data Sources 16
        - 8.1.1 Internal Sources 16
        - 8.1.2 External Sources 16

- 8.2 Data Ingestion 16
    - 8.2.1 Tools Used 16
    - 8.2.2 Update Frequency 16
- 8.3 Data Quality Controls 17
- 8.4 Data Transformation 17
- 8.5 Analytics Processing 17
- 8.6 Reporting Delivery 17
- 8.7 Business Users 17

9. DATA QUALITY MANAGEMENT FRAMEWORK 18
    - 9.1 Overview 18
    - 9.2 Data Quality Dimensions 18
    - 9.3 Data Quality Check Types 18
        - 9.3.1 Completeness Checks 18
        - 9.3.2 Accuracy Checks 19
        - 9.3.3 Consistency Checks 19
        - 9.3.4 Timeliness Checks 19
        - 9.3.5 Uniqueness Checks 20
        - 9.3.6 Validity Checks 20
    - 9.4 Data Quality Rules by Domain 20
    - 9.5 Data Quality Monitoring and Reporting 21
    - 9.6 Data Quality Issue Resolution Process 21
    - 9.7 Data Quality Tools and Technologies 22

10. DATA FUNCTIONS OVERVIEW 23
    - 10.1 Three Main Categories of Data Functions 23
        - 10.1.1 Data Services 23
        - 10.1.2 Data Governance 23
        - 10.1.3 Data Science / AI 23
    - 10.2 Key Digital Functions Related to Data 23

- 10.3 Job Categories in Data Functions 24

11. GLOBAL DATA MANAGEMENT (GDM) 27

- 11.1 Responsibilities 27
- 11.2 Roles and Governance Structure 27
  - 11.2.1 Global Level Roles 27
  - 11.2.2 Site/Local Level Roles 27
- 11.3 Data Content Owners by Domain 28
- 11.4 Data Governance and Quality 29
  - 11.4.1 Data Stewardship Model 29
  - 11.4.2 Data Quality Targets 29
  - 11.4.3 Metadata Management (Informatica CDGC) 29

12. STAKEHOLDER MAPS AND TABLES 30

- 12.1 Internal Data Providers 30
  - 12.1.1 Sales & Field Force 30
  - 12.1.2 Global Business Units (GBUs) 30
  - 12.1.3 Finance & Controlling 30
  - 12.1.4 Business Operations & Support 31
- 12.2 External Data Providers 31
- 12.3 Digital & Technology Teams 31
- 12.4 Data Consumers from I&A 31

13. KPIs AND PERFORMANCE MANAGEMENT 32

- 13.1 Data Pipeline KPIs 32
- 13.2 Business Impact Metrics 32

14. KEY SYSTEMS FOR I&A 32

- 14.1 Core Systems 32
- 14.2 Supporting Systems 33

1. SANOFI OVERVIEW

1.1 Mission

Chasing the Miracles of Science

1.2 Organizational Structure at Sanofi

1.2.1 Global Business Units (GBUs)

Sanofi is organized with three main Global Business Units (GBUs) that integrate global franchises and country-level commercial and medical organizations:

1. Specialty Care
    - Focuses on Dupixent and specialty products for:
        - Rare Diseases
        - Multiple Sclerosis
        - Oncology & Immunology
2. General Medicine
    - Handles products such as:
        - Insulins
        - Clexane
        - New product launches
3. Vaccines
    - Dedicated to Sanofi's vaccine portfolio

1.2.2 Sanofi Global Functions

Global Function	Definition
Corporate Affairs	Manages external communications, public relations, government affairs, and corporate reputation to enhance Sanofi's public image and stakeholder relationships.
Finance	Oversees financial planning, accounting, reporting, budgeting, and financial controls to ensure fiscal responsibility and support business decision-making.

Digital	Drives digital transformation, technology strategy, data management, and IT infrastructure to enable innovation and operational efficiency across the organization.
Global Business Operations	Provides operational excellence support, process optimization, and cross-functional coordination to streamline business operations and drive efficiency.
Global Manufacturing & Supply	Manages worldwide manufacturing operations, supply chain, quality assurance, and product distribution to ensure reliable delivery of medicines to patients.
Global Research & Development	Conducts scientific research, drug discovery, clinical trials, and product development to create innovative treatments and advance medical science.
Global Partnering & Business Development	Identifies and manages strategic partnerships, licensing deals, acquisitions, and external collaborations to expand Sanofi's capabilities and market reach.
People & Culture	Manages human resources, talent acquisition, employee development, organizational culture, and workplace policies to attract, retain, and develop top talent.
Internal Audit & Risk Management	Provides independent assurance, risk assessment, compliance monitoring, and internal controls to protect the organization and ensure regulatory adherence.
Legal, Compliance & Business Integrity	Ensures legal compliance, manages litigation, oversees regulatory matters, and maintains ethical business practices across all operations and jurisdictions.

1.3 Core Therapeutic Areas

Immunology

- Focus: Inflammatory diseases and dermatology
- Key Product: Dupixent

Neurology

- Focus: Neurological disorders
- Pipeline: Multiple candidates

Oncology

- Focus: Cancer treatments

- Key Product: Sarclisa

Rare Diseases

- Focus: Genetic and rare conditions
- Key Products: Altuvoct, Fitusiran

Vaccines

- Focus: Preventive immunization
- Pipeline: Rabies prevention and broad portfolio

General Medicines

- Focus: Cardiovascular and diabetes care
- Key Products: Cardiovascular portfolio, diabetes care

2. CORPORATE STRUCTURE

2.1 Executive Committee

Chief Executive Officer: Belen, Garijo (Incoming 17-Feb-26)

2.2 Major Divisions

- Global Business Units
- Corporate Functions
- Business Operations (EVP: Madeleine Roach)

3. GO-TO-MARKET CAPABILITIES (GTMC) UNDER BUSINESS OPERATIONS

3.1 GTMC Executive Leadership

Head of Commercial Excellence GTMC: Quynh Szuba

3.2 GTMC Hyderabad Hub Structure

3.2.1 Insights & Analytics (I&A) - Hub

Head of I&A Capabilities Hub: Abhishek Thapliyal

I&A Team Leads:

- Immuno/DUPI: Akansha Singh
- Global - MRCI: Sanchit Dahikar

- Global - Business Analytics: Syed Mohammad Shadab
- Global - Forecasting: Janak Raj Singh Rajput
- Global - Data Management: Vishwajeet Paul
- US - Business Analytics: Abhishek Gupta
- US - Data Management: Rhit Srivastav

3.2.2 Customer Facing Operations (CFC) - Hub

Head of Customer Facing Operations: Isabel Viramontes

CFC Team Leads:

- Customer Facing Effectiveness (Group Lead): Umang Srivastava
- Customer Facing CRM & Platforms Hub (Hub Lead): Rahul Upadhyay

3.2.3 Omnichannel & LEX/MEX - Hub

Head of Omnichannel/LexMex Commercial Excellence GTMC Hub: Archana Rai

Omnichannel Team Leads:

- HCP Campaign Operations: Pankaj Bagul
- Content Operations: Nikhil Haradh
- GenAI: Shiva Rachamadugu
- Launch & Commercialization Excellence: Yogesh Kamble
- Web and Social Media Operations: Sushant Shirke
- Capabilities and Operations: Ramnath Nayak
- Content Operations: Atulya Nagarsenkar

4. INSIGHTS AND ANALYTICS (I&A)

4.1 Overview

Insights and Analytics (I&A) is a core pillar within GTMC responsible for transforming raw data into actionable business intelligence. I&A enables data-driven decision-making across commercial operations by providing analytics, reporting, and insights to support sales, marketing, and strategic planning.

4.2 I&A Core Teams

4.2.1 Business Insights and Analytics (BI&A)

- Delivers business intelligence and analytical insights to commercial teams
- Creates performance reports, trend analysis, KPI tracking for business reviews
- Produces dashboards, reports, and presentations using Power BI, QlikSense, Excel

4.2.2 Commercial Data Management

- Manages and maintains commercial data quality and integrity
- Performs data cleansing, master data alignment, validation, and governance
- Ensures clean and reliable datasets using Snowflake, Informatica, SAP

4.2.3 Visual Analytics and Data Visualization

- Creates visual representations of data for easy understanding
- Develops dashboards, designs reports, and builds interactive visualizations
- Uses Power BI, QlikSense, and Tableau for visual reporting

4.2.4 Advanced Data Analytics and Processing Technology (ADAPT)

- Regional commercial data platform for standardized analytics
- Manages data foundations, KPI standardization, and data processing
- Delivers harmonized commercial data and regional analytics using Snowflake, Informatica CDI, dbt

4.2.5 Sales Force Effectiveness (SFE) Analytics

- Measures and optimizes sales force performance
- Analyzes call frequency, territory coverage, calculates incentives, tracks performance
- Produces SFE reports and territory optimization using Veeva CRM, Bsmart, QlikSense

4.2.6 Market Intelligence and Competitive Analysis

- Provides market insights and competitive intelligence
- Conducts market sizing, competitor tracking, market share analysis
- Uses IQVIA data (MIDAS, panels), syndicated reports, primary research

4.2.7 Customer Analytics

- Understands customer behavior and optimizes engagement

- Performs customer segmentation, journey mapping, engagement analysis
- Creates targeting lists and recommendations using Veeva CRM, Turing, Power BI

4.2.8 Forecasting and Predictive Analytics

- Predicts future trends and supports planning
- Develops demand forecasting, sales predictions, scenario modeling
- Uses Databricks, Snowflake, Python for advanced analytics

4.3 I&A Value Proposition

- Faster Decision-Making: Real-time insights enable quick business decisions
- Data-Driven Strategy: Analytics support evidence-based commercial strategies
- Operational Efficiency: Automated reporting reduces manual effort
- Competitive Advantage: Market intelligence helps stay ahead of competitors
- Revenue Growth: Insights drive sales optimization and market share gains

5. SYSTEMS, DATABASES, AND PLATFORMS

5.1 Enterprise Databases

5.1.1 On-premise

- Oracle
- PostgreSQL
- MySQL
- Microsoft SQL Server

5.1.2 Cloud databases

- AWS RDS: Oracle, SQL Server, PostgreSQL
- Azure: Oracle VM, SQL Server, PostgreSQL, MySQL
- Tencent Cloud (China): SQL Server RDS, PostgreSQL RDS, MySQL RDS

5.1.3 Specialized systems

- SAP/ERP 6.0 with Oracle databases
- AWS DynamoDB
- Data warehousing solutions

5.2 Operational and Commercial Systems

- SAP ERP and iShift (S/4HANA + MDG)
- Veeva CRM (OneCRM)
- Bsmart
- KAM Tool
- Modern Trade Dashboard
- Secondary Sales Tool
- Omnipresence
- TRIMM
- Shubhlabh

5.3 System Data Sources

System	Data Provided
SAP ERP	Primary sales transactions, Customer master data, Product master data, Financial transactions, Order data
Veeva CRM	HCP profiles, Call reports, Territory assignments, Coaching reports
BSmart	Sales-In data, Incentive data, Performance tracking
Omnipresence	Effort data, Activity tracking, Grievance management
KAM Tool	Key account data, Hospital sales data, Contract data
TRIMM	Activity data, Event tracking, Engagement metrics
Shubhlabh	Channel sales effort data, KPI data, Distribution metrics
Modern Trade Dashboard	Modern trade sales data, Retail performance, Adjustment data
Secondary Sales Tool	Sales-Out data, Sell-through metrics, Inventory movement

6. DATA FOUNDATIONS

6.1 ADAPT (Regional Commercial Data and Analytics Platform)

6.1.1 Definition

Analytics Data and Performance Technology - Regional platforms built on Snowflake to standardize commercial analytics and KPIs per market or cluster.

6.1.2 Architecture

- Data Foundation: Snowflake-based (raw, cleansed, analytical layers)
- Data Processing: Informatica CDI, dbt, Snowflake tasks, orchestrated via Control-M or Airflow
- Analytics & Reporting: Power BI dashboards for sales performance, market share, territory achievement, promotional analytics, brand performance

6.1.3 Regional Variants

- ADAPT International – Africa
- ADAPT International – Brazil
- ADAPT International – CSE (Singapore)
- ADAPT International – Eurasia
- France Commercial Data and Analytics Platform
- GSA Commercial Data and Analytics Platform
- Iberia Commercial Data and Analytics Platform
- Italy Commercial Data and Analytics Platform
- NE (Nebraska) Commercial Data and Analytics Platform
- UK and IE (Ireland) Commercial Data and Analytics Platform

6.1.4 Data Domains

- Sales and Forecast Data: IQVIA national/subnational market data, promotional audits, MIDAS
- Internal Sales Data: SAP sales transactions
- CRM Data: Veeva CRM interactions and alignments
- Patient Data: Patient and program data
- Promotional Data: Investment and campaign data
- Trade Data: Distribution and channel data

6.1.5 Benefits

- Harmonized KPIs (Key Performance Indicator) and business logic

- Reduced manual reporting effort
- Faster time to insight
- Cross-country comparability and roll-ups

6.2 MSDF (Market and Sales Data Foundation)

6.2.1 Purpose

Integrates IQVIA and non-IQVIA sources with internal sales, promotion, and patient program data.

6.2.2 Output

- Feeds Market Explorer dashboards
- Supports global analyses and reporting

6.3 Advanced Analytics Platforms

6.3.1 AWS

- Specific use cases and data exploration
- Serverless query service

7. TYPES OF DATA MANAGED BY GTMC AND I&A

7.1 Commercial Data

- Sales Performance: Revenue, units sold, growth rates
- Market Share: Brand positioning vs competitors
- Customer Data: HCP profiles, accounts, relationships
- Territory Data: Geographic performance, coverage metrics

7.2 Market Intelligence Data

- Market Sizing: Total addressable market, segment sizes
- Growth Analysis: Market trends, expansion opportunities
- Competitor Intelligence: Competitive positioning, share analysis
- Promotional Effectiveness: Campaign performance, ROI metrics

7.3 Customer Insights Data

- Segmentation: HCP categorization, behavioral groups

- Journey Mapping: Customer touchpoint analysis
- Engagement Metrics: Interaction frequency, channel preferences

7.4 Operational Data

- Sales Force Effectiveness (SFE): Call frequency, territory coverage
- Channel Performance: Distribution effectiveness, reach metrics
- Supply Metrics: Inventory levels, availability tracking

7.5 Financial Data

- Revenue Analytics: Performance tracking, trend analysis
- Pricing Analysis: Price optimization, elasticity studies
- Budget Management: Actual vs planned performance
- Discounts & Rebates: Trade spend effectiveness

8. END-TO-END DATA FLOW AND OPERATING MODEL

8.1 Data Sources

8.1.1 Internal Sources

- SAP ERP: Primary sales, secondary sales, financial data
- Veeva CRM: Customer interactions, calls, territory alignment
- Bsmart: Sales-in data and incentives tracking
- Other Systems: KAM, Modern Trade, Secondary Sales tools

8.1.2 External Sources

- IQVIA: Market data panels, MIDAS global database, promotional audits
- Master Data: OneProduct (product data), OneCustomer (customer data)
- Metadata: Informatica CDGC for business glossary and data lineage

8.2 Data Ingestion

8.2.1 Tools Used

- Informatica IICS
- Control-M and Airflow for scheduling
- dbt for transformations

- Snowflake tasks

8.2.2 Update Frequency

- SAP data: Every night
- Veeva CRM: Every four hours
- Bsmart: Daily morning update
- IQVIA: Weekly or monthly based on contracts

8.3 Data Quality Controls

- Completeness and accuracy checks
- Data consistency validation
- Master data matching (products, customers)
- Automated alerts for data issues

8.4 Data Transformation

- Apply business rules to raw data
- Standardize country, product, and customer codes
- Create unified master records
- Calculate KPIs like market share and growth

8.5 Analytics Processing

- Create aggregated data views
- Build market performance models
- Develop sales force effectiveness metrics
- Create forecasting and predictive models

8.6 Reporting Delivery

- Power BI dashboards
- QlikSense for sales tracking
- Automated reports (daily, weekly, monthly)
- Self-service data access for analysts

8.7 Business Users

- GTMC teams across regions

- Sales management and field teams
- Finance and Market Access departments
- Regional and country leadership

9. DATA QUALITY MANAGEMENT FRAMEWORK

9.1 Overview

Data Quality Management is a critical component of Sanofi's data ecosystem, ensuring that all data used for decision-making is accurate, complete, consistent, and reliable. The framework establishes standards, processes, and controls to maintain high-quality data across all systems and platforms.

9.2 Data Quality Dimensions

Dimension	Definition	Business Impact
Completeness	All required data fields are populated with valid values	Incomplete data leads to inaccurate reporting and missed insights
Accuracy	Data correctly represents the real-world entity or event	Inaccurate data results in wrong business decisions
Consistency	Data values are uniform across all systems and databases	Inconsistent data causes confusion and reconciliation issues
Timeliness	Data is available when needed and reflects current state	Delayed data impacts real-time decision-making
Uniqueness	No duplicate records exist in the dataset	Duplicates inflate metrics and cause operational inefficiencies
Validity	Data conforms to defined formats, ranges, and business rules	Invalid data causes system errors and processing failures

9.3 Data Quality Check Types

9.3.1 Completeness Checks

Check Type	Description	Example

Null Value Check	Identifies missing or null values in required fields	Customer ID cannot be null
Mandatory Field Check	Ensures all required fields are populated	Product code, sales date, quantity must be filled
Record Count Check	Validates expected number of records received	Daily sales file should have minimum 1000 records
Coverage Check	Ensures data covers all expected dimensions	All territories should have sales data

9.3.2 Accuracy Checks

Check Type	Description	Example
Range Validation	Values fall within acceptable ranges	Sales quantity between 1 and 10,000
Cross-Reference Check	Data matches against master data sources	Product codes match OneProduct master
Historical Comparison	Current values compared to historical patterns	Sales variance within 50% of previous period
Source-Target Reconciliation	Data matches between source and target systems	SAP totals match Snowflake totals

9.3.3 Consistency Checks

Check Type	Description	Example
Cross-System Consistency	Same data values across different systems	Customer name same in CRM and ERP
Referential Integrity	Foreign keys match primary keys in related tables	Territory ID exists in territory master
Format Consistency	Data follows standard formats	Date format YYYY-MM-DD across all sources
Business Rule Consistency	Data adheres to defined business rules	Discount cannot exceed 40% of list price

9.3.4 Timeliness Checks

Check Type	Description	Example

Data Freshness Check	Data is updated within expected timeframe	CRM data refreshed within 4 hours
SLA Monitoring	Data delivery meets service level agreements	Daily reports available by 8 AM
Latency Check	Time between data creation and availability	Sales transactions available within 24 hours
Timestamp Validation	Data timestamps are current and valid	No future-dated transactions

9.3.5 Uniqueness Checks

Check Type	Description	Example
Duplicate Detection	Identifies duplicate records	No duplicate customer IDs
Primary Key Validation	Ensures unique primary keys	Each transaction has unique ID
Fuzzy Matching	Identifies potential duplicates with similar values	Similar customer names flagged for review
Cross-Source Deduplication	Removes duplicates across multiple sources	Single customer record from multiple systems

9.3.6 Validity Checks

Check Type	Description	Example
Format Validation	Data matches expected format	Phone numbers in correct format
Domain Validation	Values belong to valid domain list	Country codes from approved list
Data Type Validation	Data matches expected data type	Numeric fields contain only numbers
Business Logic Validation	Data passes business rule validation	Order date before ship date

9.4 Data Quality Rules by Domain

Domain	Key Data Elements	Quality Rules
Sales Data	Transaction ID, Product Code, Customer ID, Quantity, Amount	No nulls in key fields, Amount > 0, Valid product codes

Customer Data	Customer ID, Name, Address, Specialty, Territory	Unique customer ID, Valid territory assignment
Product Data	Product Code, Name, Brand, Therapeutic Area	IDMP compliant, Valid brand mapping
Territory Data	Territory ID, Region, Country, Sales Rep	Complete hierarchy, No orphan territories
Market Data	Market Code, Period, Volume, Value, Share	Valid market codes, Share between 0-100%

9.5 Data Quality Monitoring and Reporting

9.5.1 Data Quality Dashboard

The Data Quality Dashboard provides real-time visibility into data quality metrics across all domains:

Metric	Description	Target	Frequency
Overall DQ Score	Composite score across all dimensions	Above 95%	Daily
Completeness Rate	Percentage of complete records	Above 98%	Daily
Accuracy Rate	Percentage of accurate records	Above 99%	Daily
Timeliness Rate	Percentage of on-time data delivery	Above 95%	Daily
Issue Resolution Time	Average time to resolve DQ issues	Under 24 hours	Weekly

9.5.2 Data Quality Reports

Report Name	Description	Audience	Frequency
Daily DQ Summary	Overview of daily data quality metrics	Data Stewards	Daily
Weekly DQ Trend Report	Week-over-week quality trends	Data Management Leads	Weekly
Monthly DQ Scorecard	Comprehensive monthly quality assessment	Leadership	Monthly
Exception Report	Detailed list of data quality failures	Technical Teams	Daily

Root Cause Analysis Report	Analysis of recurring quality issues	Data Governance	As needed

9.6 Data Quality Issue Resolution Process

Step 1: Issue Detection

- Automated quality checks identify issues
- Alerts generated for critical failures
- Issues logged in tracking system

Step 2: Issue Classification

- Severity: Critical, High, Medium, Low
- Category: Completeness, Accuracy, Consistency, Timeliness
- Source: System, Process, Human Error

Step 3: Issue Assignment

- Assigned to appropriate Data Steward
- Escalation path defined based on severity
- SLA clock starts

Step 4: Root Cause Analysis

- Investigate source of issue
- Document findings
- Identify systemic vs. one-time issues

Step 5: Resolution

- Implement fix in source system
- Correct affected data
- Validate resolution

Step 6: Prevention

- Update quality rules if needed
- Implement additional controls
- Document lessons learned

9.7 Data Quality Tools and Technologies

Tool	Purpose	Usage
Informatica Data Quality	Data profiling, cleansing, and standardization	Enterprise-wide data quality
Informatica CDGC	Data cataloging and governance	Metadata management
Snowflake Data Quality	In-database quality checks	Analytics platform quality
Great Expectations	Automated data validation	Pipeline quality checks
Custom SQL Scripts	Specific business rule validation	Domain-specific checks
Power BI DQ Dashboards	Quality visualization and reporting	Stakeholder reporting

10. DATA FUNCTIONS IN SQL

10.1 Overview

SQL (Structured Query Language) provides a comprehensive set of built-in functions to manipulate, analyze, and transform data within databases. These functions are essential tools for data professionals at Sanofi to extract insights, ensure data quality, and support business intelligence operations.

10.2 Aggregate Functions

Aggregate functions perform calculations on multiple rows and return a single summary value.

Function	Description	Example Usage
COUNT()	Counts the number of rows or non-null values	SELECT COUNT(*) FROM sales;
SUM()	Calculates the total sum of numeric values	SELECT SUM(revenue) FROM transactions;
AVG()	Calculates the average of numeric values	SELECT AVG(price) FROM products;
MIN()	Returns the minimum value	SELECT MIN(order_date) FROM orders;

MAX()	Returns the maximum value	SELECT MAX(salary) FROM employees;

10.3 String Functions

String functions manipulate and transform text data.

Function	Description	Example Usage
CONCAT()	Combines multiple strings into one	SELECT CONCAT(first_name, ' ', last_name) FROM customers;
LENGTH() / LEN()	Returns the length of a string	SELECT LENGTH(product_name) FROM products;
UPPER()	Converts string to uppercase	SELECT UPPER(country) FROM locations;
LOWER()	Converts string to lowercase	SELECT LOWER(email) FROM users;
SUBSTRING()	Extracts part of a string	SELECT SUBSTRING(phone, 1, 3) AS area_code FROM contacts;
TRIM()	Removes leading/trailing spaces	SELECT TRIM(customer_name) FROM customers;
REPLACE()	Replaces characters in a string	SELECT REPLACE(phone, '-', '') FROM contacts;
LEFT() / RIGHT()	Extracts characters from left/right	SELECT LEFT(product_code, 3) FROM products;

10.4 Date and Time Functions

Date functions handle temporal data operations.

Function	Description	Example Usage
GETDATE() / NOW()	Returns current date and time	SELECT GETDATE() AS current_datetime;
YEAR()	Extracts year from date	SELECT YEAR(order_date) FROM orders;

MONTH()	Extracts month from date	SELECT MONTH(hire_date) FROM employees;
DAY()	Extracts day from date	SELECT DAY(transaction_date) FROM transactions;
DATEDIFF()	Calculates difference between dates	SELECT DATEDIFF(day, start_date, end_date) FROM projects;
DATEADD()	Adds interval to date	SELECT DATEADD(month, 3, order_date) AS due_date FROM orders;
DATEPART()	Extracts specific part of date	SELECT DATEPART(quarter, sales_date) FROM sales;

10.5 Mathematical Functions

Mathematical functions perform numeric calculations.

Function	Description	Example Usage
ROUND()	Rounds to specified decimal places	SELECT ROUND(price, 2) FROM products;
CEILING() / CEIL()	Rounds up to nearest integer	SELECT CEILING(discount_rate) FROM promotions;
FLOOR()	Rounds down to nearest integer	SELECT FLOOR(tax_amount) FROM invoices;
ABS()	Returns absolute value	SELECT ABS(profit_loss) FROM financial_data;
POWER()	Raises number to power	SELECT POWER(growth_rate, 2) FROM metrics;
SQRT()	Returns square root	SELECT SQRT(variance) FROM statistics;
MOD()	Returns remainder of division	SELECT MOD(employee_id, 10) FROM employees;

10.6 Conditional Functions

Conditional functions implement logic within queries.

Function	Description	Example Usage
CASE WHEN	Conditional logic expression	SELECT CASE WHEN salary > 50000 THEN 'High' ELSE 'Low' END FROM employees;
COALESCE()	Returns first non-null value	SELECT COALESCE(mobile, phone, 'No contact') FROM customers;
NULLIF()	Returns NULL if values are equal	SELECT NULLIF(discount, 0) FROM products;
ISNULL() / IFNULL()	Replaces NULL with specified value	SELECT ISNULL(commission, 0) FROM sales_reps;
IIF()	Simple if-then-else (SQL Server)	SELECT IIF(stock > 0, 'Available', 'Out of Stock') FROM inventory;

10.7 Window Functions (Analytical Functions)

Window functions perform calculations across rows related to the current row.

Function	Description	Example Usage
ROW_NUMBER()	Assigns unique sequential number	SELECT ROW_NUMBER() OVER (ORDER BY salary DESC) FROM employees;
RANK()	Assigns rank with gaps for ties	SELECT RANK() OVER (ORDER BY sales DESC) FROM sales_data;
DENSE_RANK()	Assigns rank without gaps	SELECT DENSE_RANK() OVER (ORDER BY score DESC) FROM performance;
LAG()	Accesses previous row value	SELECT LAG(sales, 1) OVER (ORDER BY month) FROM monthly_sales;
LEAD()	Accesses next row value	SELECT LEAD(sales, 1) OVER (ORDER BY month) FROM monthly_sales;
PARTITION BY	Divides result set into partitions	SELECT AVG(salary) OVER (PARTITION BY department) FROM employees;

10.8 Conversion Functions

Conversion functions change data from one type to another.

Function	Description	Example Usage

CAST()	Converts data type	SELECT CAST(price AS DECIMAL(10,2)) FROM products;
CONVERT()	Converts with format options	SELECT CONVERT(VARCHAR, order_date, 101) FROM orders;
TRY_CAST()	Safe conversion (returns NULL on failure)	SELECT TRY_CAST(text_field AS INT) FROM data_table;
FORMAT()	Formats values with patterns	SELECT FORMAT(order_date, 'yyyy-MM-dd') FROM orders;

11. GLOBAL DATA MANAGEMENT (GDM)

11.1 Responsibilities

- Master Data Governance: Oversight of core business data
- Reference Data Management: Managing standardized codes and values
- Data Quality Management: Ensuring data accuracy and completeness
- Metadata and Lineage Management: Tracking data origins and transformations
- Compliance and Regulatory Controls: GxP, IDMP/ISO standards, ALCOA+ principles

11.2 Roles and Governance Structure

11.2.1 Global Level Roles

Global Data Steward

- Reviews basic data modifications
- Manages shared material types across sites
- Reviews file uploads for data changes
- Ensures global data consistency

Global GxP Final Approver

- Validates GxP-critical data
- Approves regulatory data changes
- Ensures compliance with pharmaceutical standards
- Final authority on quality-critical data

11.2.2 Site/Local Level Roles

Data Process Manager (DPM)

- Ensures data quality at site level
- Manages data timeliness
- Controls data maturity levels
- Oversees site-level governance

Data Steward

- Tracks data accuracy
- Oversees data lifecycle management
- Facilitates data accessibility
- Provides training to data users

Data Content Owner (DCO)

- Sole accountable person for master data
- Defines data values and business rules
- Documents calculations and logic
- Ensures compliance with business requirements

GxP Final Approver (Local)

- Validates GxP data at site level
- Approves data creation and changes
- Validates change controls
- Ensures regulatory compliance

11.3 Data Content Owners by Domain

Domain	DCO Role
HSE (Health, Safety, Environment)	Data Domain Material Expert (HSE)
Market	Data Domain Material Expert (Market)
Planning	Data Domain Material Expert (Planning)
Purchasing	Data Domain Material Expert (Purchasing)

Quality Management	Data Domain Material Expert (Quality Management)
Warehouse Management	Data Domain Material Expert (Warehouse Management)
Business/Finance	Data Requester Material (Business_Finance)

11.4 Data Governance and Quality

11.4.1 Data Stewardship Model

- Clear ownership and accountability structure
- Defined roles at global and local levels
- Continuous monitoring and improvement

11.4.2 Data Quality Targets

Metric	Target
Data Completeness	Above 95%
Data Accuracy	Above 98%
Timeliness (Critical Data)	Under 4 hours
Master Data Consistency	100%

11.4.3 Metadata Management (Informatica CDGC)

Technical Metadata Catalog

- Technical names and structures
- Database schemas and tables
- System connections and interfaces

Business Glossary

- Business terms and definitions
- Standardized terminology
- Data element descriptions

Lineage Tracking

- Full audit trail of data transformations
- Source-to-target mapping

- Impact analysis capabilities

Data Classification

- Critical Data Element (CDE) identification
- Sensitivity classification (DERC framework)
- Data ownership assignment

12. STAKEHOLDER MAPS AND TABLES

12.1 Internal Data Providers

12.1.1 Sales & Field Force

Stakeholder	Data Provided
Field Force (FF)	Customer call data, visit reports, sample distribution
Area Managers (AM)	Territory performance, team activity reports
Regional Business Managers (RBM)	Regional sales data, market coverage, competitive intelligence
Key Account Managers (KAM)	Key account sales, hospital data, contract performance
Channel Sales Managers (CSM)	Channel performance, distribution metrics

12.1.2 Global Business Units (GBUs)

Stakeholder	Data Provided
Specialty Care	Product sales targets, brand strategy, launch plans
General Medicine (GenMed)	Product performance targets, market access data, pricing
Vaccines	Vaccination campaign data, seasonal demand forecasts
Business Unit Heads (BUH)	Strategic priorities, budget allocations

12.1.3 Finance & Controlling

Stakeholder	Data Provided
Zonal Finance Managers (ZFM)	Revenue data, budget vs actual reports
Controlling Team	Cost center data, profitability analysis, financial KPIs

12.1.4 Business Operations & Support

Stakeholder	Data Provided
Sales Force Effectiveness (SFE)	Incentive data, performance metrics, territory alignment
BI&A Team	Processed analytics, dashboard data, report outputs

12.2 External Data Providers

Vendor	Data Provided
IQVIA	MIDAS global market data, national/sub-national sales, promotional audits, prescription data, competitive intelligence
Veeva Systems	CRM platform data, customer master data, call activity, territory alignment
Other Research Agencies	Primary market research, ad-hoc studies, customer surveys

12.3 Digital & Technology Teams

Team	Role in I&A
Snowflake COE	Data warehouse management, database provisioning, technical support
Enterprise Data Governance	Data governance policies, metadata management (CDGC), data quality standards
Data Engineering Team	ETL pipeline development, data integration (Informatica), transformations
OneMesh Team	Data marketplace management, data product cataloging
OneAI Team	AI/ML model development, predictive analytics

12.4 Data Consumers from I&A

Consumer	Data/Insights Received
Executive Leadership	Executive dashboards, strategic KPIs, market summaries
GBU Leadership	Product performance, market share, competitive intelligence
Country/MCO Leadership	Country dashboards, territory analytics, local insights
Sales Management	Sales performance reports, territory optimization, customer segmentation
Marketing Teams	Campaign effectiveness, customer insights, brand performance
Finance Teams	Revenue analytics, profitability analysis, forecasting data
Strategy Teams	Market intelligence, competitive analysis, growth opportunities

13. KPIs AND PERFORMANCE MANAGEMENT

13.1 Data Pipeline KPIs

Metric	Target
Data Processing Time	Under 4 hours (end-to-end)
Data Quality Score	Above 95%
System Availability	Above 99.5%
Data Freshness (Critical Data)	Under 4 hours
User Satisfaction	Above 90%

13.2 Business Impact Metrics

- Faster decision-making across commercial teams
- Increased data-driven commercial decisions
- Reduced manual reporting effort and costs
- Revenue and market share improvement from insights

14. KEY SYSTEMS FOR I&A

14.1 Core Systems

System	Purpose	Relevance to I&A
OneProduct	Product master data management (IDMP-compliant)	Product data for analytics and reporting
OneCustomer	Customer master data with 360-degree profiles	Customer segmentation and insights
iShift (SAP S/4HANA)	Core business transactions and master data	Sales data, financial data, order data

14.2 Supporting Systems

System	Purpose	Relevance to I&A
LUNA (LUCY)	Laboratory information management	Quality and compliance data integration
Tender Management System	Tender lifecycle and bidding management	Contract and pricing analytics